SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Exp :2 Write the Python Program to solve 8-Queen Problem.

Aim:

To write the Python Program to solve 8-Queen Problem

Algorithm:

Step1: Start with an empty board of size 8x8.

Step2: Place the first queen in the first row and the first column.

Step3: For each subsequent row, check all possible columns to place the queen, and backtrack if a valid position cannot be found.

Step4: A valid position is a column where no other queens are in the same row, column, or diagonal.

Step5: If no solution is found, return failure.

Program:

```
def is_safe(board, row, col):
    for i in range(row):
        if board[i][col] == 1:
            return False
    i = row
    j = col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    i = row
    j = col
    while i >= 0 and j < 8:
```

```
            if board[i][j] == 1:
                return False
            i -= 1
            j += 1

    return True

def solve_n_queens(board, row):
    if row >= 8:
        return True

    for col in range(8):
        if is_safe(board, row, col):
            board[row][col] = 1

            if solve_n_queens(board, row + 1):
                return True

            board[row][col] = 0

    return False

def print_board(board):
    for row in board:
        line = ""
        for cell in row:
            if cell == 1:
                line += "Q "
            else:
                line += ". "
        print(line)
```

```
if __name__ == "__main__":
    chess_board = [[0 for i in range(8)] for j in range(8)]

    if solve_n_queens(chess_board, 0):
        print_board(chess_board)
    else:
        print("Failure")
```

OUTPUT:

```
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```

Result: Hence, the simple python program for 8-Queen Problem is done